PROJECT SPECIFICATION & REQUIREMENT DOCUMENT

ENG-AI001: Legacy Dataset Profiling & Ingestion Pipeline

Project Stream: Knowledge Graph Foundations	**Priority:** Medium
Target Audience: Engineering Interns	**Date:** June 2026

📜 Project Background

OntoGraphica has been engaged to modernize a client's legacy data infrastructure. As a critical first step, our team has been granted access to an uncurated archive containing historical system and personnel records in flat CSV format.

Before this unstructured data can be ingested into our core **Knowledge Graph Engine** and transformed into semantic triples, we must establish a rigorous data profiling baseline. This task requires developing a local data-validation utility to audit the target file for structural integrity, identify missing data fields, and generate an automated diagnostic report.

🛠️ Core Engineering Directive: Responsible AI Usage

Strict Policy Regarding Generative AI Tools

- **Minimize Code Generation:** Do not use AI to write, copy-paste, or generate the actual lines of code for your script.
- **Use AI to Learn, Not to Prompt:** Use AI exclusively as a conceptual mentor. If you encounter an obstacle, use it to understand underlying mechanics (e.g., *"How does pandas handle encoding issues internally?"* or *"Explain standard deviation calculation on missing data"*), then write the code yourself.
- **The Goal:** Building a strong foundational understanding of data parsing mechanics now will prepare you for complex semantic mapping architectures later.

🎯 Functional Objectives

You are required to build a modular Python utility (`csv_analyzer.py`) that programmatically executes the following data engineering operations:

1. **Data Ingestion:** Load the source CSV file safely, handling standard encoding variations and edge-case delimiters.

2. **Structural Profiling:** Calculate total row counts (representing target entities) and column counts (representing potential properties).
3. **Schema Inspection:** Extract and catalog all field names to map out the explicit schema header.
4. **Data Quality Audit:** Scan the dataset to detect, count, and locate missing or null values across all attributes.
5. **Statistical Evaluation:** Compute baseline descriptive statistics for numerical data (means, ranges) and frequency distributions for categorical data.
6. **Automated Reporting:** Programmatically write all analytical outputs to a structured text file named `analysis_report.txt`.

✅ Success & Acceptance Criteria

To complete this first task successfully, your script and report must satisfy the following straightforward technical requirements:

- ☐ **Error-Free Execution:** Running the program must load the file and run to completion without throwing any console errors or crashing.
- ☐ **Row & Column Verification:** The script must correctly count the exact number of rows and columns present in the provided CSV file.
- ☐ **Null Value Count:** The output must display the exact number of missing/empty cells found in each column.
- ☐ **Matching Output File:** Executing the script must automatically create a new file named `analysis_report.txt` in the same directory, containing the counts and statistics.

🎁 Next-Phase Progression

Downstream Integration: Successful delivery of this module unlocks project **ENG-KG001: CSV-to-RDF Semantic Mapping Engine**, where you will convert this validated flat data into a functional knowledge graph.

📦 Technical Submission Checklist

- ☐ `csv_analyzer.py` *(Core Python utility)*
- ☐ `analysis_report.txt` *(Automated pipeline output)*
- ☐ `README.md` *(Setup instructions, dependency lists, and architectural notes)*
- ☐ `proof.png` *(A CLI screenshot verifying flawless script execution and runtime)*

📖 Glossary of Engineering Terms

To assist with onboarding, use the following industry standard definitions for the semantic web concepts applied in this project:

Flat Data / Spreadsheets

Data constrained to two-dimensional tables (rows and columns) where relationships between different datasets are implicit rather than machine-readable or natively linked.

Knowledge Graph (KG)

A relational network of real-world entities (objects, individuals, concepts) and their semantic relationships, explicitly defined as an interconnected web of data.

Entity

A distinct, uniquely identifiable node in a domain (e.g., a specific employee, project, or department). In flat data, a single row typically maps to an entity.

Attribute / Property

A contextual characteristic or data type that describes or connects an entity. In flat data, these are represented by column headers.

Ingestion

The automated process of extracting, validating, and importing raw data from an external file source into a software application for processing.

RDF (Resource Description Framework)

The universal standard model for semantic data exchange on the Web, structuring statements as machine-readable "triples" consisting of a Subject, Predicate, and Object.

Semantic Transformation

The engineering process of converting standard, unstructured, or tabular datasets into a schema-governed, relationship-first format (like RDF) to preserve contextual meaning.